

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

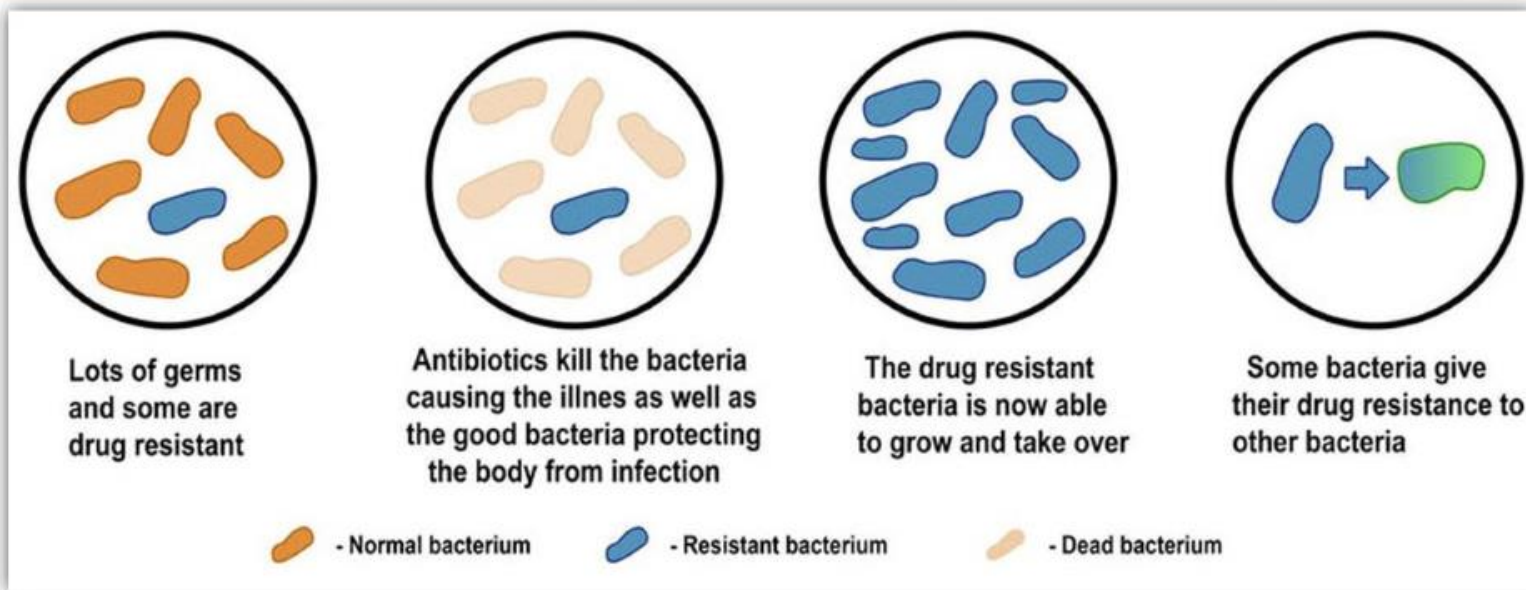
MODULE - 19

BIOTECHNOLOGY - 9

ANTIMICROBIAL RESISTANCE

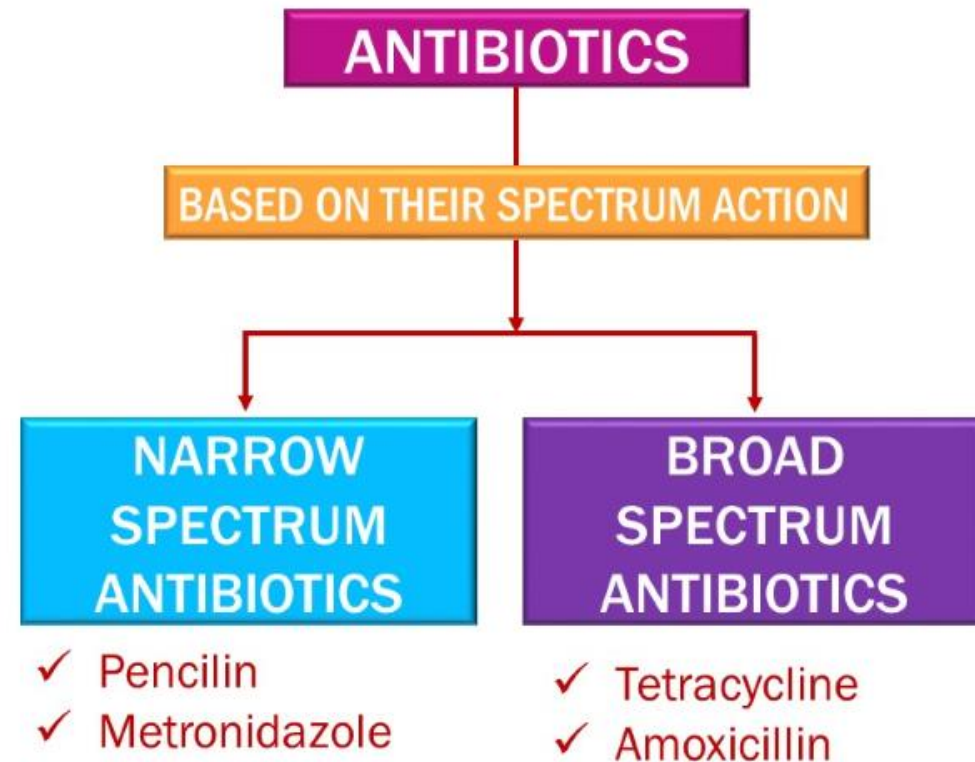
Antimicrobials are Antibiotics, Antivirals, Antifungals and Antiprotozoal drugs.

- Antimicrobial resistance happens when microorganisms such as bacteria, viruses and fungi evolve when they are exposed to antimicrobial agents such as antibiotics, antivirals and antifungals and develop resistance to them.
- As a result, the medicines become ineffective and infection persists in the body.
- Microorganisms that develop antimicrobial resistance are called 'drug resistant microbes' or 'super bugs.'



ANTIBIOTICS

- Antibiotics are used in the treatment and prevention of bacterial infections. Antibiotics may either kill the bacteria or inhibit their growth.
- Antibiotics can be broadly classified as
 - a) Narrow Spectrum Antibiotics:**
These are antibiotics that target only specific kinds of bacteria.
 - b) Broad Spectrum Antibiotics:**
These are used to treat wide range of bacterial infections.



USE OF ANTIBIOTICS

- They are used to treat or prevent bacterial infections and to a limited extent protozoal infections.
- Antibiotics are also given as preventive medicine in patients with weakened immune system.
- They are administered before performing a surgery.
- **Animal Husbandry** – Antibiotics are widely used as growth supplements in livestock to improve the overall health of animals for producing higher quality and quantity of milk and meat.



WHY IS ANTIMICROBIAL RESISTANCE A GLOBAL CONCERN?

1. Common and simple infectious diseases become difficult to treat resulting in prolonged illness, disability and even death.
2. Without effective antimicrobial drugs, procedures such as surgeries, organ transplants and diabetes management become very complicated and risky.
3. Antimicrobial resistance increases the cost of health care with lengthier stay in hospital and intensive care.



CAUSES OF ANTIMICROBIAL RESISTANCE

1. Inappropriate use of antibiotics

The main reason for resistance to antibiotics is their misuse –

- a) Unnecessary prescription of antibiotics for viral infections.
- b) Frequent prescription of broad-spectrum antibiotics rather than narrow spectrum ones.
- c) Failure of patients to adhere to the course prescribed by doctors, and
- d) Self-medication.

2. Indiscriminate use of antibiotics in agriculture and animal husbandry

- a) Antibiotics are sprayed onto fruit and vegetable plants to prevent and treat infections. This may lead to emergence of resistant strains of bacteria.
- b) Antibiotics are widely overused in animal husbandry and veterinary practice.



CAUSES OF ANTIMICROBIAL RESISTANCE

3. Insufficient Research and Development

- a) Antibiotics are not as profitable as drugs that treat chronic conditions such as diabetes and psychiatric disorders, as they are used for short duration. So, pharma companies do not consider R&D into antibiotics an economically wise investment.

4. Regulatory Lapses

- a) Over the counter availability of antibiotics in India is one the biggest reason behind antibiotic misuse in India.



ANTIMICROBIAL RESISTANCE

Fighting antimicrobial resistance

- a) Better hygiene – preventing infections from happening in the first place.
- b) Having access to clean drinking water.
- c) Vaccination for all.

WHAT IS ANTIMICROBIAL RESISTANCE (AMR)?

Antimicrobials?

Substances used to treat a wide variety of infectious diseases in humans and animals. They:

- kill micro-organisms
- stop micro-organisms from growing and multiplying

Example: antibiotics



Antimicrobial resistance?

The ability of micro-organisms to withstand antimicrobial treatments.

Example: MRSA (meticillin-resistant *Staphylococcus aureus*) commonly present on human skin and mucous membranes



Why is resistance growing?

- Overuse of antibiotics
- Misuse of antibiotics
- Spread through various routes



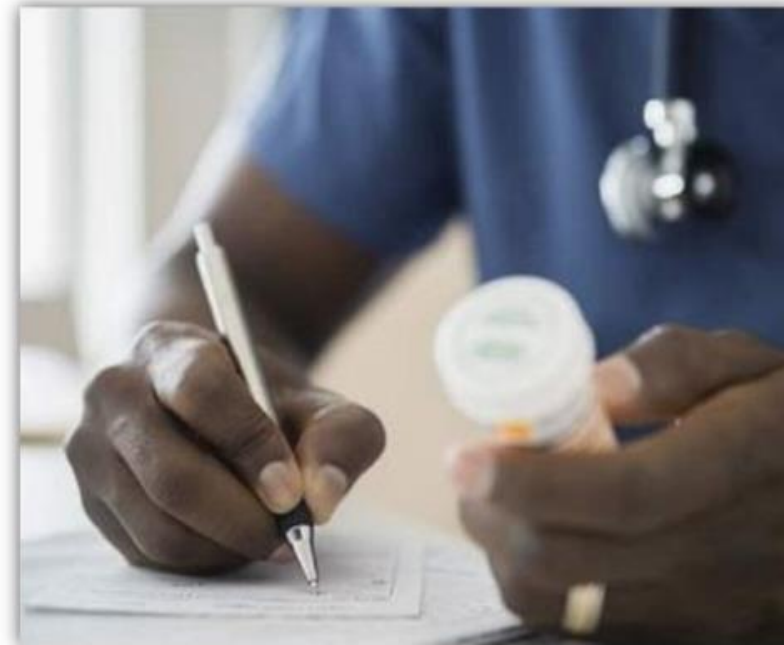
Effect of growing resistance?

- Treatment may become ineffective
- Serious risk to public health

FIGHTING ANTIMICROBIAL RESISTANCE

RESPONSIBILITIES OF PEOPLE

- a) Using antibiotics only when prescribed by the doctor.
- b) Completing the full antibiotic course, even if they feel better.
- c) Not sharing antibiotics with others.
- d) Stopping the use of antibiotics as growth promoters in farm animals.



FIGHTING ANTIMICROBIAL RESISTANCE

RESPONSIBILITIES OF HEALTHCARE WORKERS

- a) Prescribing antibiotics only when they are truly needed.
- b) Prescribing the right antibiotics to treat the infections (rather than broad spectrum antibiotics).
- c) Adhering to the policies set by the government.



FIGHTING ANTIMICROBIAL RESISTANCE

RESPONSIBILITIES OF POLICYMAKERS

- a) Banning over-the counter sale of antibiotics.
- b) Having national plans and guidelines for combating AMR.
- c) Encouraging industries to carry out more R&D to produce better drugs.



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